Pro-Sport Complex Management: An IoT-Based and Smart Booking Solution

SWR302 Course Project – Class SE20A02 – Group 3

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No Institute Given

Abstract. The abstract should briefly summarize the contents of the paper in 150–250 words. It should clearly state the problem statement, the proposed system architectural design, and the key outcomes of the project.

Keywords: Smart Booking · Social Matching · Escrow · Software Engineering.

1 Introduction

Introduction to your project goes here[cite: 1]. You can talk about the practical problems in sports complex management, booking conflicts, and issues with traditional court booking systems[cite: 1].

2 Project Team and Responsibilities

In this section, we present the project team structure and the assignment of tasks among the members of Group 3 (Class SE20A02) for the SWR302 course.

Table 1. Project Team Members and Contribution Roles.

No	Student ID	Full Name	GitHub Username	Role	Main Responsibility
1	DE190408	Hồ Võ Minh Vỹ	minhvy259	Leader	Project Management / Backend
2	DE190147	Phạm Nguyễn Tiến Đạt	tiendat2277sg-wq	Member	Backend / Database
3	DE190900	Dương Khang Huy	khuy17	Member	Frontend / UI Design
4	DE191003	Hồ Tấn Phát	Phat1425	Member	Frontend Developer
5	DE190130	Nguyễn Đăng Phúc	chuotphuc311005-lgtm	Member	Software Tester / QA

3 Related Work

Review existing systems or platforms in the market. To cite a document from your BibTeX file, use the cite command like this [1].

4 System Architecture and Methodology

Describe the system workflow here[cite: 1]. You can include details about the multi-tier architecture:

- **Frontend:** Developed using ReactJS, Tailwind CSS, and Shadcn UI[cite: 1].
- **Backend:** Implemented with Java Web, Servlets, JSP, and Hibernate/JPA[cite: 1].
- **Database:** Managed by Microsoft SQL Server[cite: 1].

4.1 Smart Booking and Escrow Mechanism

Detail the concurrency control logic for court booking and the escrow mechanism for matching players without financial risks[cite: 1].

5 Results and Discussions

Present your dashboards, experiment results, or system evaluation data here[cite: 1].

Table 2. System Performance under Concurrent Booking Requests.

Metrics	**Traditional System**	**Our Proposed System**
Conflict Rate (%)	12.4	0.0
Response Time (ms)	450	120

6 Conclusion and Future Work

Summarize the main contributions of your software development project and potential enhancements for the SWR302 course project, such as mobile application deployment or advanced analytics dashboards.

References

1. Author, A., Author, B.: Title of the journal paper. Journal Name **Volume**(Issue), Pages–Pages (Year)
2. Author, C.: Title of the published book. Publisher, City (Year)